

Arrow Technology Expo – Technical Class Descriptions

Thursday, October 22, 2026
8:30am to 3:00pm

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Session #1 - 8:30am to 9:30am

Advanced Energy

Power Supply System EMI Fundamentals

As power electronic systems continue to operate at higher switching frequencies and power densities, electromagnetic interference (EMI) has become a critical design consideration affecting product performance, regulatory compliance, reliability, and time-to-market. This training session provides a practical introduction to EMI concepts with a focus on power supply applications and real-world mitigation techniques.

The session begins with an overview of EMI fundamentals, explaining what EMI is, how it is generated, and why reducing EMI is essential for modern electronic systems. Participants will gain an understanding of the electric and magnetic field mechanisms that create electromagnetic noise and how these fields couple into surrounding systems.

Building on these fundamentals, the course examines EMI behavior within power supply systems, covering both conducted and radiated emissions. Special attention is given to the dominant noise paths found in switching power supplies, and the mechanisms by which these disturbances propagate through equipment and system interconnections.

The training also reviews key EMI compliance standards and regulatory requirements commonly encountered in industrial, medical, and commercial applications. Participants will learn how emissions are measured, interpreted, and evaluated against applicable standards.

The session will include an overview of Advanced Energy's EMI resources, facilities, and support capabilities, including laboratory infrastructure, compliance engineering expertise, and field and customer support.

Digi International

Edge AI in Practice: Designing Intelligent Connected Devices with the Digi ConnectCore 95

Modern embedded products demand on-device intelligence, robust connectivity, and built-in security — all without sacrificing reliability or long-term maintainability. This session explores how the Digi ConnectCore 95, powered by the NXP i.MX 95, delivers on all fronts: up to six Arm Cortex-A55 cores, an integrated NPU for edge AI/ML, Wi-Fi 6E, multi-gigabit Ethernet with TSN, and industrial-grade thermal performance — wrapped in Digi's TrustFence security framework and backed by cloud-based OTA and lifecycle security services.

Whether you're designing for industrial automation, medical devices, or connected infrastructure, this session gives you a practical foundation for evaluating the ConnectCore 95 in your next design.

Microchip Technology

Design Faster, Control Smarter: Unlock High-Performance Embedded Control with dsPIC® Digital Signal Controllers

As embedded applications demand higher performance, faster control loops, and greater system efficiency, Microchip Technology's dsPIC® Digital Signal Controllers provide a compelling platform for real-time, closed-loop control design. This seminar explores how dsPIC DSCs help address complex engineering challenges in motor control, power conversion, and high-frequency sensing applications. Through an instructor-led demonstration using the dsPIC Curiosity Nano platform, Building Block Solution Kits, and MPLAB® X IDE, attendees will gain insight into how Microchip's development ecosystem simplifies prototyping, accelerates design cycles, and supports faster market adoption.

Silicon Labs

Enable Accurate Distance Estimation Using Channel Sounding

Explore the latest advancements in Bluetooth 6, with a special focus on Channel Sounding. Channel Sounding is a new standardized, interoperable, and secure method for distance measurement that is set to revolutionize a variety of applications. This session will also feature a comparative analysis of Channel Sounding with other ranging methods.

Phoenix Contact

Overcoming connectivity challenges for seamless robotic integration

Robotics relies on robust connectivity for real-time control, data exchange, and system integration. This encompasses internal electrical and electronic connections as well as communication between robots and external devices. Performance is often constrained by latency, bandwidth limitations, and network instability—particularly in autonomous vehicles, industrial automation, and multi-robot environments. Advances in connector design and emerging technologies offer improvements, yet achieving reliable, low-latency connectivity in dynamic conditions remains a critical challenge for fully autonomous and collaborative robotics.

PUI Audio, Inc.

PUI Audio: Elevating UX Design with Audio, Haptics, & Sensors

Discover how PUI Audio's audio, haptic, and sensor technologies enhance user experience in modern product design - from smart devices to medical wearables. Learn how integrating sound, tactile feedback, and environmental sensing improves interaction and functionality. This session features real-world applications and practical strategies for designing intuitive, responsive, and engaging products.

Session #2 - 9:40am to 10:40am

Lattice Semi

Utilizing FPGA's as Companion chips in embedded applications

As MCU/MPU demands grow in embedded and edge AI applications, designers can offload tasks to small, low-power, low-latency FPGAs. This seminar shows how adding FPGAs expands system headroom and enables new features without requiring major changes to existing processors from Qualcomm, NXP, ST, Nvidia, and others

Molex

Innovation Driven by Automotive, Designed for Rugged Applications

The challenges facing today's automotive industry—higher power, faster data, smaller packaging, and uncompromising reliability—are accelerating innovations that extend well beyond passenger vehicles. As electrification, software-defined architectures, automotive Ethernet, and advanced sensing become mainstream, connector technologies are evolving to deliver greater performance in increasingly demanding environments.

The seminar will showcase a broad range of Molex automotive technologies that are increasingly finding applications beyond passenger vehicles. Whether you're designing commercial vehicles, industrial equipment, agricultural machinery, robotics, marine systems, or other rugged applications, you'll discover how technologies developed for the automotive industry's demanding requirements can help improve reliability, performance, and enable scalable architectures to prepare your products for the next generation of connectivity.

onsemi

Power Protection Reimagined – Leveraging onsemi's eFuse Portfolio for Next-Generation System Reliability

As power density, system complexity, and uptime requirements continue to increase, intelligent power protection is becoming essential across industrial, automotive, and computing applications. This training explores the growing adoption of eFuses and highlights how onsemi's power protection portfolio delivers faster, more accurate fault protection, enhanced diagnostics, and improved system reliability compared to traditional protection methods.

Qualcomm

Qualcomm: Moving AI to the Edge while being more developer friendly

Qualcomm is advancing edge AI by enabling intelligent processing directly on devices, addressing critical requirements around latency, privacy, power efficiency, and overall system cost. This approach leverages heterogeneous compute architectures—spanning CPU, GPU, and NPU—to match the right workload to the right engine while supporting real-world use cases such as vision, robotics, industrial automation, and IoT. At the same time, Qualcomm is prioritizing a more developer-friendly experience by reducing friction across the AI lifecycle, from model selection and optimization to deployment and fleet management. By closing the gap between proof-of-concept demos and scalable production

systems, Qualcomm's edge AI ecosystem allows developers and partners to move faster, deploy securely, and bring AI-enabled products to market with greater confidence.

Skyworks

Connectivity Isn't One Size Fits All: The Tradeoffs Engineers Should Consider in Industrial Networks

Should you deploy Wi-Fi, Bluetooth, LoRaWAN, Wi-SUN, or private 5G? The answer is often "it depends." As industrial systems become more connected, engineers are increasingly challenged to balance competing requirements such as range, throughput, reliability, battery life, and infrastructure costs. Selecting the wrong connectivity strategy can lead to performance limitations, higher operating costs, and difficult upgrades down the road.

In this session, we'll cut through the hype and examine the real engineering tradeoffs behind today's leading wireless technologies. Using practical industrial use cases, you'll learn how to evaluate connectivity options, avoid common design pitfalls, and build networks that are ready for both current and future demands.

TDK

Powering the Shift to Solid-State Transformers: How TDK Technologies Enable the Transition

The move to solid state transformers (SSTs) requires compact, high frequency power architectures and robust component solutions. This session describes how TDK component technologies support SST design at both the component and system levels.

Key topics:

- Component solutions: advanced magnetics, DC link and CeraLink capacitors, high voltage contactors, circuit protection, and sensing modules.
- Performance benefits: enabling higher switching frequencies, increased power density, improved thermal management, and enhanced safety and reliability.
- Enablers and trends: impact of wide bandgap devices (SiC, GaN) on SST requirements and the role of materials innovation in reducing size and weight while improving lifetime.
- Practical guidance: component selection tradeoffs, layout and thermal considerations, and system level integration strategies to accelerate SST deployment.

Intended for power electronics and system engineers seeking concise, product focused guidance on designing scalable, reliable solid state transformer solutions using TDK technologies.

Session #3 - 10:50am to 11:50am

Bourns

Hardware That Works When Nothing Else Does — Integrating Bourns Passive Thermal Protection and NTC Sensing into Your Power Electronics Design

The most reliable component in your thermal architecture is one that requires no power, no firmware, and no external signal to do its job. Bourns Thermal & Precision Solutions designs passive thermal cutoff and temperature sensing components for engineers who understand that hardware protection and software monitoring are complementary layers — not interchangeable ones. The TCO mini-breaker family covers trip temperatures from 55°C to 150°C in automotive-grade, SMT, and weld-to-tab configurations, with contact ratings suited to relay coil interlocks, battery cell protection, and DC power stage applications. The NTC thermistor family spans 0201 through 0603 in resistance values from 10kΩ to 470kΩ with anti-sulfur and automotive-grade options for the most demanding sensing environments. This session gives you the technical depth to evaluate, select, and integrate both product families with confidence — application circuits, parameter tradeoffs, and real design-in examples included.

Diodes Inc

Solve common design problems with programmable, mixed-signal uASICs from Diodes Inc.

Learn how to save board space, power, and costs with Diodes' configurable uASICs solutions for industrial and consumer applications. In this session we will address common design needs like power sequencing, watchdog supervisors, waveform generators, voltage monitors, OVP/UVP and OCP functions, and the implementation of simple and complex logic functions in a programmable, mixed-signal ASIC. We will show design tool examples and demonstrate how simple it is to implement a solution using uASICs from Diodes Inc.

Ezurio

Cyber Resilience Act: Is Your Connected Product Ready?

The Cyber Resilience Act (CRA) isn't a future problem, reporting starts September 11, 2026 and most manufacturers are not ready.

This session covers what CRA actually requires: vulnerability handling, incident reporting, and the documentation you'll need for conformity assessment. Then we'll show how Ezurio builds compliance into the software stack itself with our system-on-module and wireless hardware paired with the EZ BSP and EZ Connectivity Stack. We'll close with a live look at how this comes together on real hardware: Ezurio's newest Wi-Fi 7 module, our latest SOMs, and Gateworks' Ara240 DNPU.

Honeywell

Discover How Advanced Pressure & Flow Sensing Can Transform Your Medical, Industrial and Aerospace Applications

Join us for an engaging session exploring how next generation pressure and liquid flow sensors are powering innovation across medical, industrial, and aerospace markets. These sensors play a vital role in the accuracy, safety, and reliability of critical systems—from life saving medical devices to high performance industrial and aerospace equipment.

You'll see how our sensing technologies enable smarter, more dependable operation in applications such as dialysis equipment, blood analysis systems, centrifuges, oxygen and nitrogen gas distribution, HVAC units, data storage, process controls, industrial machinery, pumps, and robotics

Kyocera AVX

Improving Industrial Controller Performance with Advanced Passive Components

Performance of Industrial Controllers is impacted by the passives around ICs that are used for power, processing, and timing. Oscillator and crystal selection impact the repeatability and accuracy of controller operations, while varistor selection for sensitive processors impact overall reliability. Timing device and circuit protection advancements and their implications for how they improve industrial controller performance will be discussed. Lastly, large capacitance values are needed for high power demands and a comparison of 3 electrolytic capacitor technologies, and their respective ESR performance will be presented.

STMicroelectronics

Powering the Next Generation of Humanoid Robots

Unlocking real-time control, edge AI, and system integration with STM32

Humanoid robots are rapidly evolving from research platforms into practical systems in different aspects of our life. As these robots become more capable, their demands on embedded processing, real-time control, sensing, connectivity, safety, and energy efficiency continue to increase. STM32 microcontrollers provide a flexible and scalable foundation to address many of these challenges, enabling developers to build responsive, power-efficient, and secure robotic systems.

During this presentation we will explore how ST's STM32 portfolio can support key functions in humanoid robots, including motion control, sensor fusion, edge AI, motor management, human-machine interaction, and system connectivity.

By combining performance, energy efficiency, and a rich ecosystem, STM32 can play a central role in enabling the next generation of humanoid robots. This talk will provide customers with a clear overview of the opportunities, technical building blocks, and practical implementation paths available with ST solutions.

Session #4 - 12:50pm to 1:50pm

Altera

Robotics, Nvidia Attach, Power and Performance advantages and design productivity

Altera 3.0 represents the next iteration of the company making it the only pure-play FPGA company in the industry today. Learn about our value proposition in Robotics, Holoscan capabilities, and our investments into our architecture and tools you need to accelerate your design flow and beat your competition to market. Our Hyperflex and Tensor Flow DSP architecture allows for higher performance, lower power, and AI at the edge. This enables you to future proof your designs with edge AI and increase your productivity with higher utilization and ease of timing closure.

Eaton Electronics

Powering Reliability: Smarter Component Choices for Efficient Designs

Learn how passive components in power systems impact performance, including thermal behavior, transients, EMI, efficiency, and lifetime. This session presents practical frameworks to size and select components, evaluate tradeoffs between cost, performance, and reliability, and identify common failure modes—enabling more robust and efficient designs.

elinfochips

EdgeAI Systems - From POCs to Production

Edge AI is moving beyond POCs to production deployments, but building scalable, reliable, and secure Edge AI systems requires more than selecting the right AI model.

This session explores the key engineering challenges of building production-ready Edge devices, highlighting the hardware platforms and software frameworks required for production-scale deployments. elinfochips will share real-world engineering experience and demonstrate how organizations can leverage System-on-Modules (SOMs) and software accelerators to reduce development complexity, shorten time-to-market, and successfully deploy Edge AI solutions across manufacturing, healthcare, transportation, and industrial automation.

Nuvoton

Designing Ultra-Low-Power Applications with the M2U51xx MCU Family

This session explores how Nuvoton's Arm® Cortex®-M23-based M2U51xx MCU family enables ultra-low-power designs. Topics include run-to-deep-power-down current, LDO and DC-DC operation, SRAM retention, autonomous low-power peripherals, wake-up options, and the M2U54 segment LCD controller with charge-pump support for metering and other low-power applications.

NXP

NXP Advanced Analog — Complete design solutions

NXP's Advanced Analog portfolio and the unique advantage it offers design teams: a single supplier capable of addressing every critical analog function across the complete system. From battery and power management — where NXP delivers scalable, safety-rated solutions spanning low-voltage IoT endpoints to high-voltage EV systems — to motor control and gate drivers that handle everything from small DC motors through high-power SiC and IGBT traction inverter applications, NXP's breadth eliminates the need to source across multiple vendors. Add in a comprehensive precision sensing and signal conditioning portfolio covering temperature, pressure, current, and universal analog front-end solutions, and NXP becomes the trusted partner for the full signal chain — from the physical world all the way to the digital processor. Attendees will come away understanding why NXP's system-solution approach, built on a foundation of functional safety, proven reliability, and scalable hardware and software, makes it the premier Advanced Analog supplier for automotive, industrial, and IoT applications.

Vishay

Vishay Optoelectronics – Ambient Light Sensing Under OLED Displays & Gate Driver Isolation

(15 minutes) Being able to sense the ambient light while under an OLED display is all about timing and each display has a unique set of variables that must be accounted for when setting our sensor's I2C directories. A brief case study and two product highlights.

(30 minutes) The Energy and Industrial markets are moving towards higher and higher operating voltages, for example, uninterruptible power supplies and motor control inverters. These applications require higher isolation. Vishay will cover three applications and schematics where optical isolation plays a key role.

Session #5 - 2:00pm to 3:00pm

Arrow

Supplier Market Update

Stay informed with Arrow Electronics as we explore the latest global market trends.

Critical Link LLC

Critical Link SOMs: Implementing AI at the Edge, Embedded Linux Security, & AMP

Building on last year's introduction to AI at the edge and AMP (Asymmetric Multiprocessing), this seminar takes a deeper look at real-world implementation. We will cover practical strategies for getting started with edge AI using Altera and Qualcomm-based SOMs, including a live Agilex 5 demo, techniques for interprocessor communication and resource partitioning across multi-processor and FPGA systems, and an attack surface analysis of embedded Linux systems.

Infineon

Machine Learning (ML) for Engineers “What is ML at the EDGE”

Machine learning enables systems to learn from data and supports better decision-making, but successful outcomes depend on a structured process. This presentation introduces the core EDGE ML; Data Collection, Preprocessing, Model Training, Model Evolution, and Model Deploy. Attendees will gain a practical understanding of how these processes work together to build reliable machine learning solutions. Through real-world examples and best practices, the session provides a clear foundation for understanding the machine learning workflow and applying its principles to solve business and organizational challenges.

Littelfuse

Smart, Safe, and Secure: Building Controls Solutions from Littelfuse

This session focuses on Littelfuse's key technologies that enhance building security and safety, covering smart locks and access control, PIR motion detection, video surveillance and doorbell systems, and smoke and heat detection.

Micro Crystal

Driving Innovation with Ultra-Low Power Timing Solutions for Key Markets

Attendees will explore the latest innovations in ultra-low power timekeeping technology and its critical role in modern electronics.

The seminar will cover:

Fundamentals of RTC Technology

How real-time clocks maintain accurate time in embedded systems and IoT devices.

Ultra-Low Power Design Principles

Techniques for minimizing energy consumption in battery-powered and energy-harvesting applications.

Integration in Emerging Applications

Use cases in wearables, medical devices, smart sensors, and industrial IoT.

Challenges and Solutions

Addressing temperature stability, long-term accuracy, and synchronization in connected environments.

Future Trends

Innovations in oscillators, energy-efficient architectures, and next-generation timing solutions.

By the end of the seminar, participants will understand how ultra-low power RTC technology enables extended battery life, reliable performance, and sustainability in cutting-edge applications.

Yageo Group / KEMET Electronics

An Investigation of Rectangular Capacitor Technology, with Advantages Compared to Traditional Cylindrical Capacitors

This seminar will provide a comprehensive overview of rectangularly shaped Aluminum Electrolytic Capacitors, with a focus on advantages versus traditional, cylindrically shaped capacitors. Rectangular capacitors have higher volumetric efficiency, provide optimized space utilization, facilitate improved thermal management through enhanced heat-sink compatibility, and outperform conventional cylindrical designs electrically in specific applications. Additionally, the impact of environmental factors, such as altitude and pressure, on thermal behavior and reliability will be discussed. The information shared is supported by comparative analyses and experimental case studies. The aim is to introduce the rectangular shape as a significant change in technology versus traditional cylindrical capacitors providing solutions for challenges experienced in electronic design. The scope of this seminar will be limited to Aluminum Electrolytic Capacitors including those with Solid Polymer Electrolyte systems.